protection. One possible protection might be the focus of public attention on him as an eminent scientist, so he searched among his scientific ideas for one that might make a big splash in West and East alike. His choice was an idea that he had been mulling over since the early 1930s: the idea that "normal" stars like the Sun might possess neutron stars at their *centers*—*neutron cores* Landau called them.

The reasoning behind Landau's idea was this: The Sun and other normal stars support themselves against the crush of their own gravity by means of thermal (heat-induced) pressure. As the Sun radiates heat and light into space, it must cool, contract, and die in about 30 million years' time—unless it has some way to replenish the heat that it loses. Since there was compelling geological evidence, in the 1920s and 1930s, that the Earth had been kept at roughly constant temperature for 1 billion years or longer, the Sun *must* be replenishing its heat somehow. Arthur Eddington and others had suggested (correctly) in the 1920s that the new heat might come from nuclear reactions, in which one kind of atomic nucleus is transmuted into another—what is now called *nuclear burning* or *nuclear fusion;* see Box 5.3. However, the details of this nuclear burning had not been worked out sufficiently, by 1937, for physicists to know whether it could do the job. Landau's neutron core provided an attractive alternative.

Just as Zwicky could imagine powering a supernova by the energy released when a normal star implodes to form a neutron star, so Landau could imagine powering the Sun and other normal stars by the energy released when its atoms, one by one, get captured onto a neutron core (Figure 5.4).